SpaceX Falcon 9 First-Stage Landing Prediction

Evidence and interpretation

Completed presentation in PDF format. This report documents the complete IBM Data Science Capstone workflow and provides substantive methods, results, visuals, interpretation, and conclusions.

Supporting visual from the completed analysis

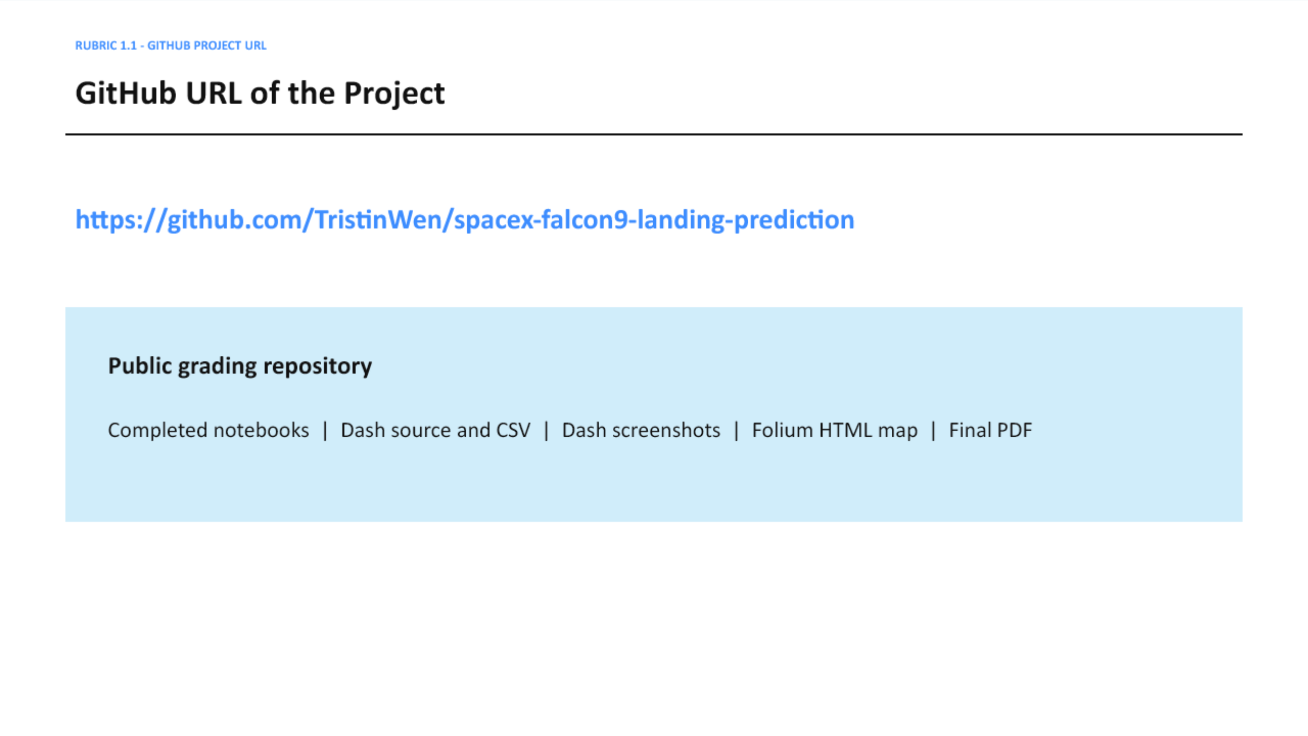
GitHub URL of the Project

Evidence and interpretation

Public repository: https://github.com/TristinWen/spacex-falcon9-landing-prediction. The repository contains seven completed notebooks, the Plotly Dash Python application and CSV data, two dashboard screenshots, an exported Folium map, and the final PDF.

Repository evidence

https://github.com/TristinWen/spacex-falcon9-landing-prediction

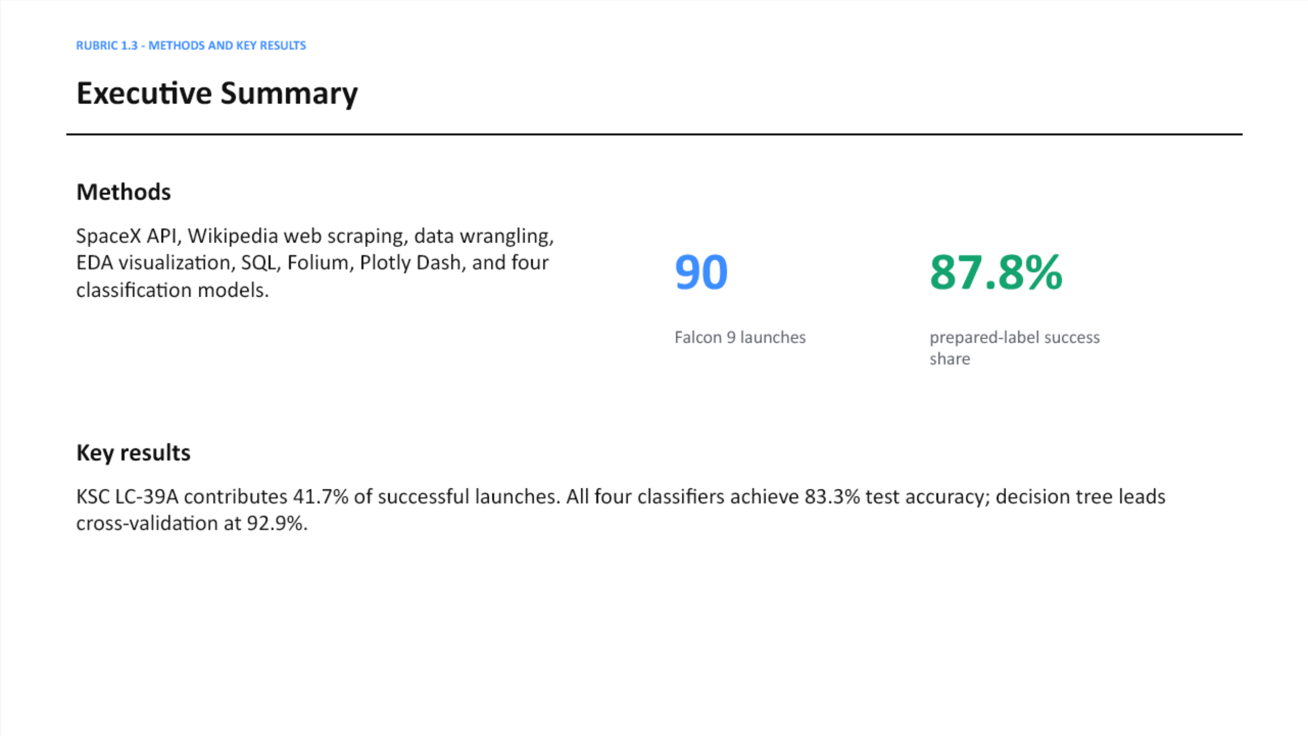


Supporting visual from the completed analysis

Executive Summary

Evidence and interpretation

The project collected Falcon 9 records through the SpaceX API and Wikipedia scraping, cleaned and explored the data with visualization and SQL, built Folium and Dash analytics, and compared four classification models. The prepared dataset contains 90 launches. KSC LC-39A contributes 41.7% of successful launches. All models reached 83.3% test accuracy, while the decision tree led cross-validation at 92.9%.

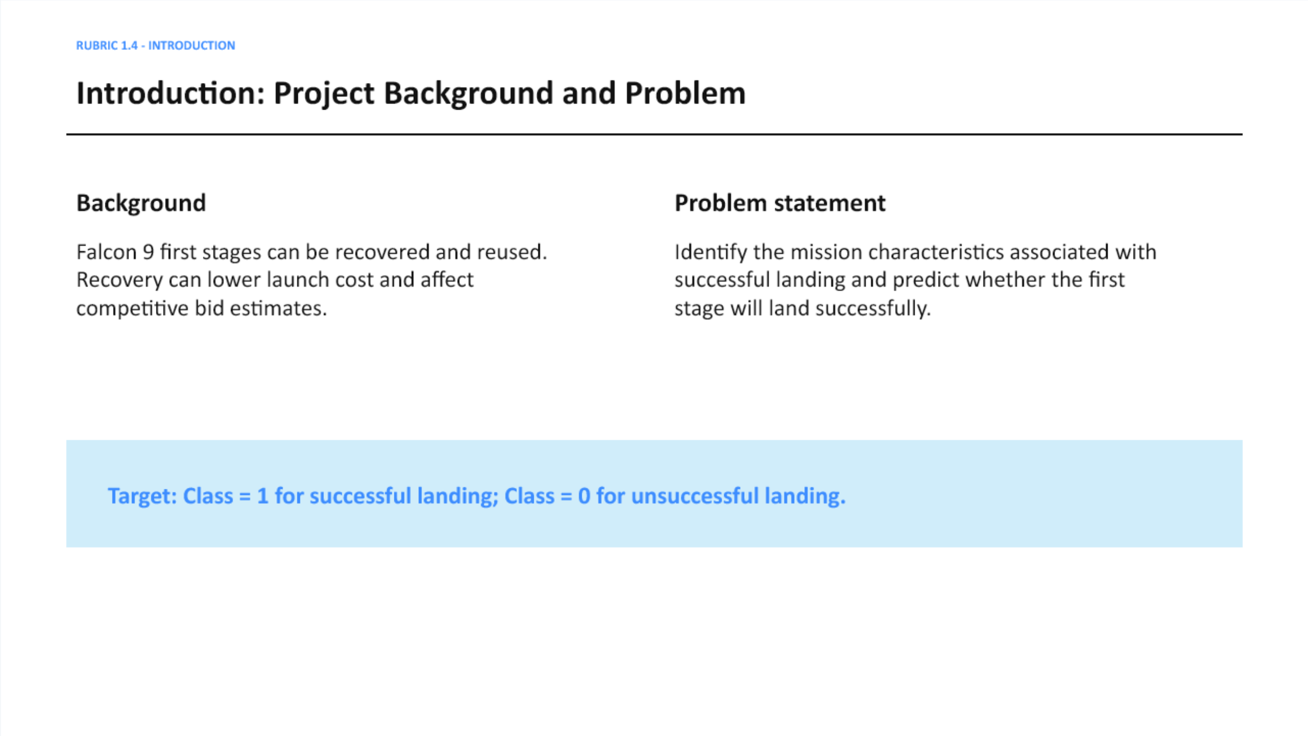


Supporting visual from the completed analysis

Introduction: Project Background and Problem

Evidence and interpretation

SpaceX can recover and reuse Falcon 9 first stages, so landing success affects launch cost and mission economics. The analysis asks which mission characteristics are associated with successful landings and whether those characteristics can predict the binary landing outcome: Class 1 for success and Class 0 for failure.



Supporting visual from the completed analysis

Data Collection - SpaceX API

Evidence and interpretation

The API workflow requested past launches, normalized JSON responses, resolved rocket, launchpad, payload, and core identifiers, assembled a pandas DataFrame, and retained Falcon 9 missions. The completed notebook produces 90 launch records and 17 analytical fields.

Repository evidence

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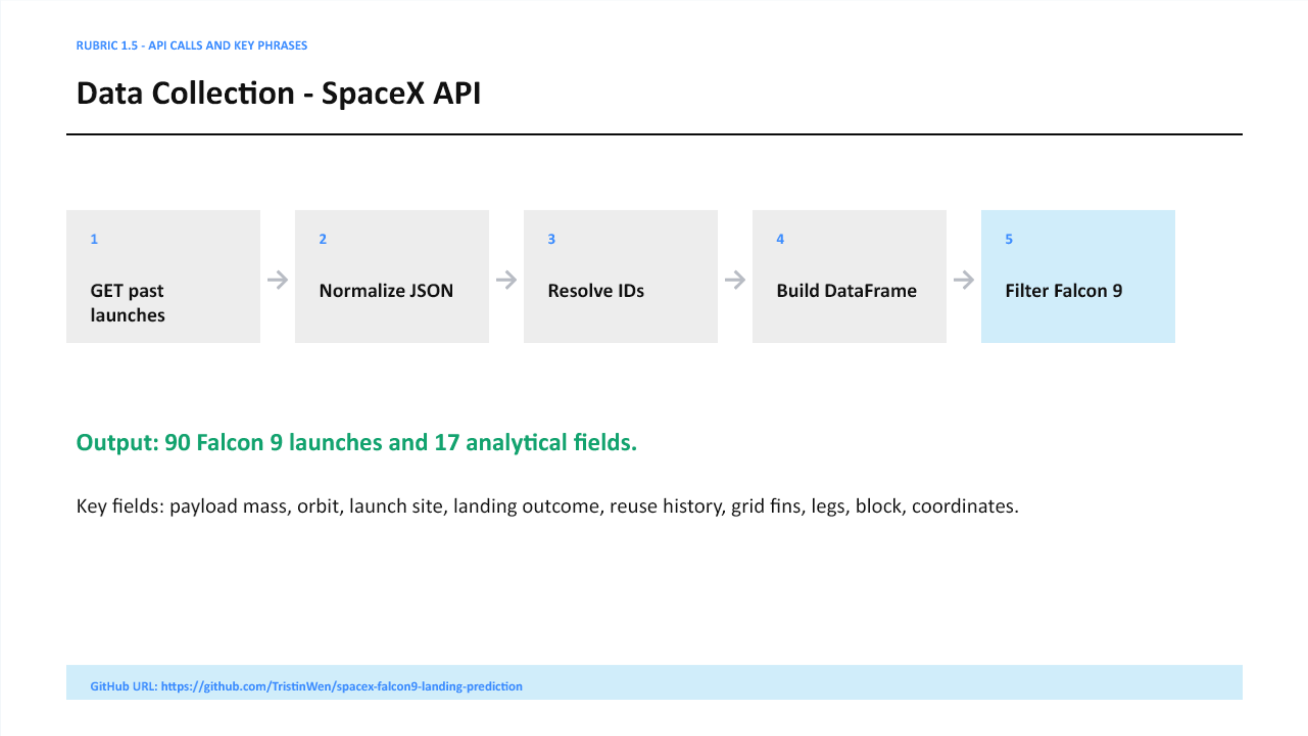

Supporting visual from the completed analysis

Data Collection - Web Scraping

Evidence and interpretation

The scraping workflow requested the historical Falcon 9 launch page, parsed the HTML with BeautifulSoup, located the launch table, extracted headers and rows, cleaned cell values, and converted the result into a pandas DataFrame containing 121 historical launch rows and 11 fields.

Repository evidence

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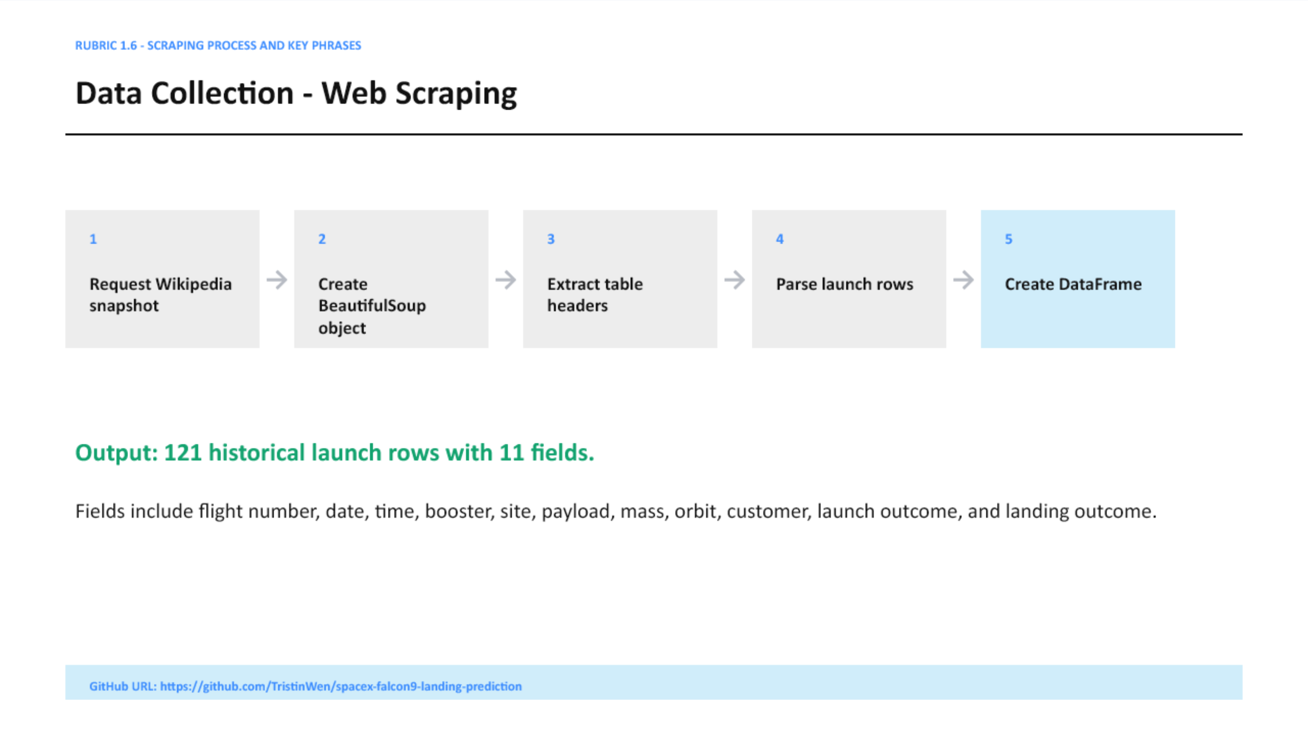

Supporting visual from the completed analysis

Data Wrangling Methodology

Evidence and interpretation

Cleaning retained the required columns, replaced missing PayloadMass values with the sample mean, preserved missing landing-pad cases, counted launches by site and orbit, mapped landing outcomes to the Class label, and produced 90 records with no missing PayloadMass values and a prepared success share of 87.8%.

Repository evidence

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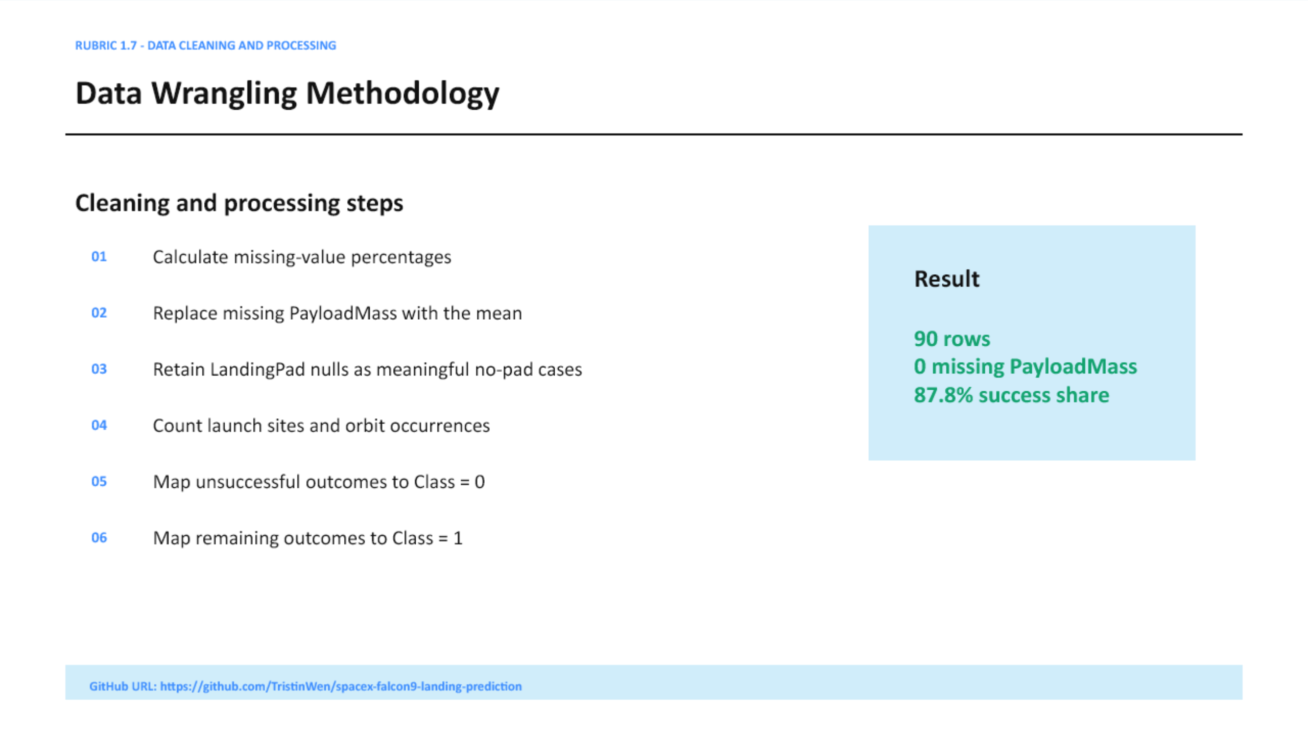


Supporting visual from the completed analysis

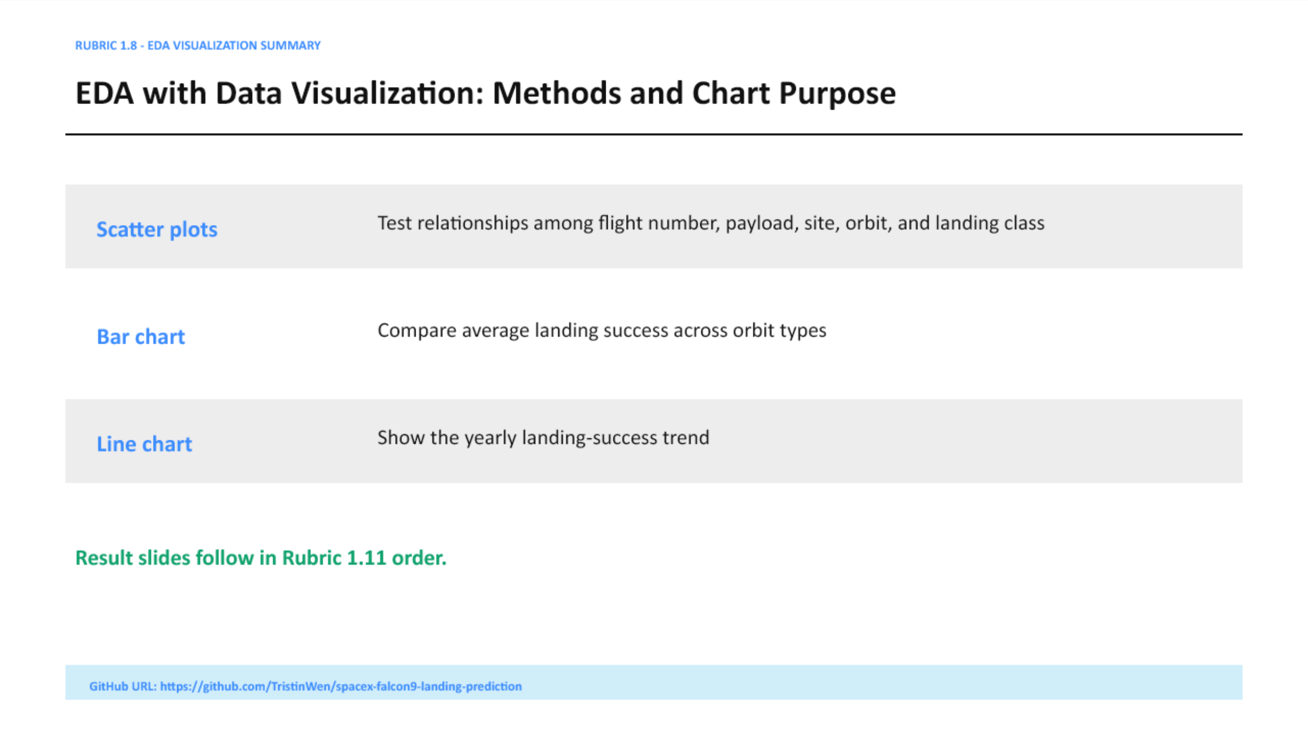
EDA with Data Visualization: Methods and Chart Purpose

Evidence and interpretation

Scatter plots test relationships among flight number, payload, launch site, orbit, and landing outcome. A bar chart compares mean success by orbit type. A line chart measures the yearly success trend. Each result slide states the chart purpose and interprets the observed pattern.

Repository evidence

https://github.com/TristinWen/spacex-falcon9-landing-prediction



Supporting visual from the completed analysis

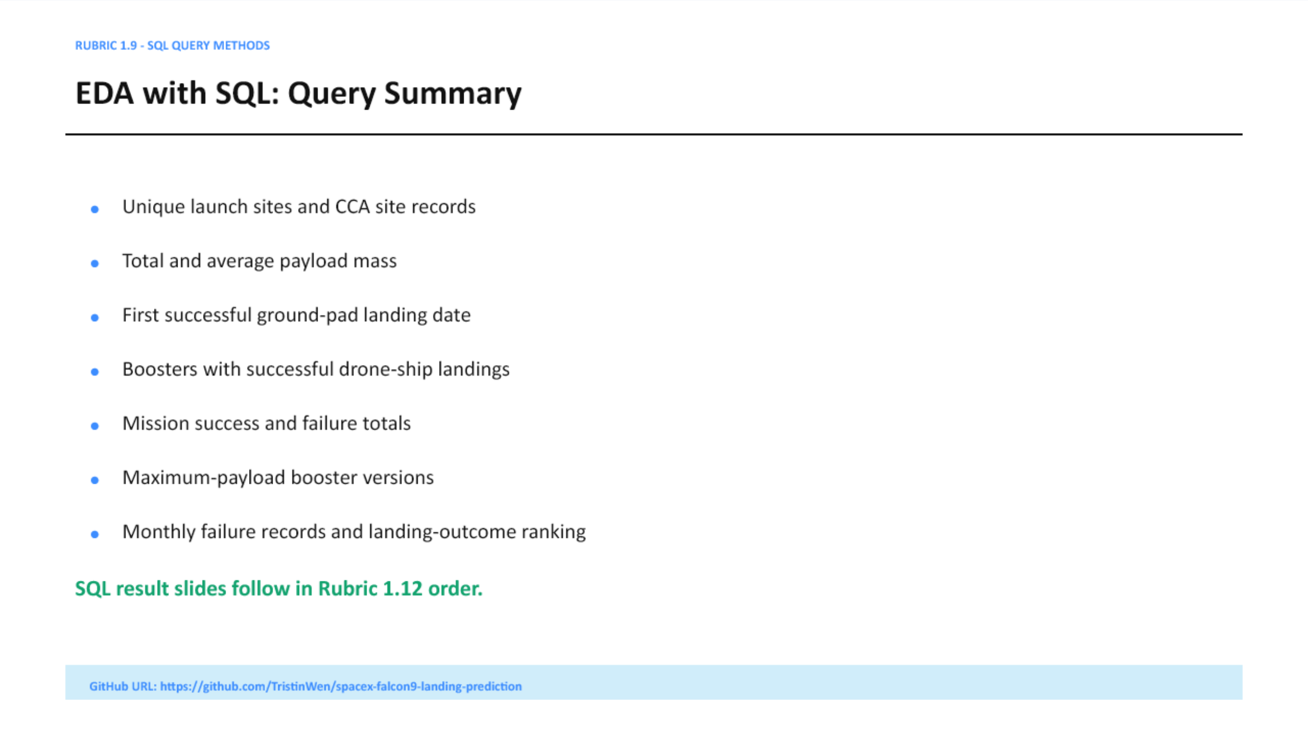
EDA with SQL: Query Summary

Evidence and interpretation

SQL queries identify unique launch sites, CCA date-range records, total and average payload mass, the first successful ground-pad landing, booster success and failure totals, maximum-payload missions, monthly failure records, and landing-outcome rankings.

Repository evidence

https://github.com/TristinWen/spacex-falcon9-landing-prediction



Supporting visual from the completed analysis

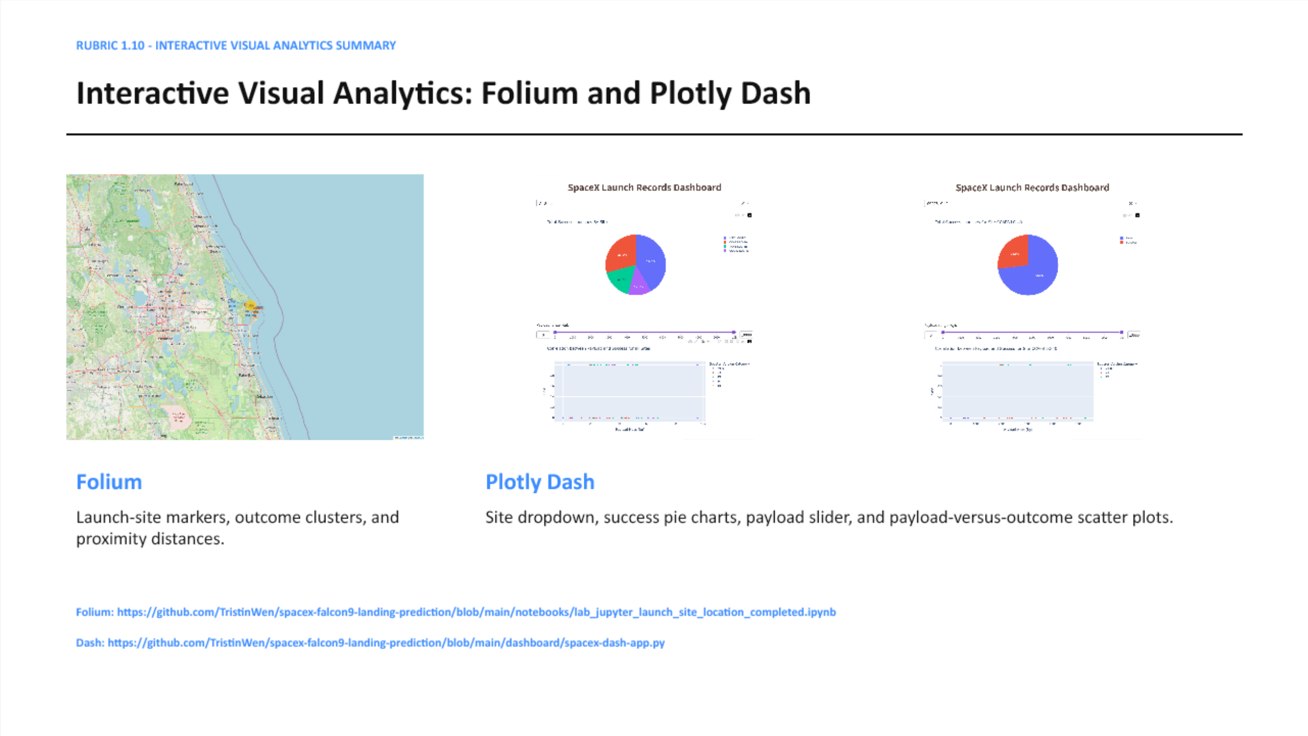
Interactive Visual Analytics: Folium and Plotly Dash

Evidence and interpretation

Folium maps launch-site coordinates, outcome markers, marker clusters, and distances to coastline, cities, highways, and railways. Plotly Dash filters records by site and payload range and dynamically updates a success pie chart and payload-versus-outcome scatter plot.

Repository evidence

https://github.com/TristinWen/spacex-falcon9-landing-prediction

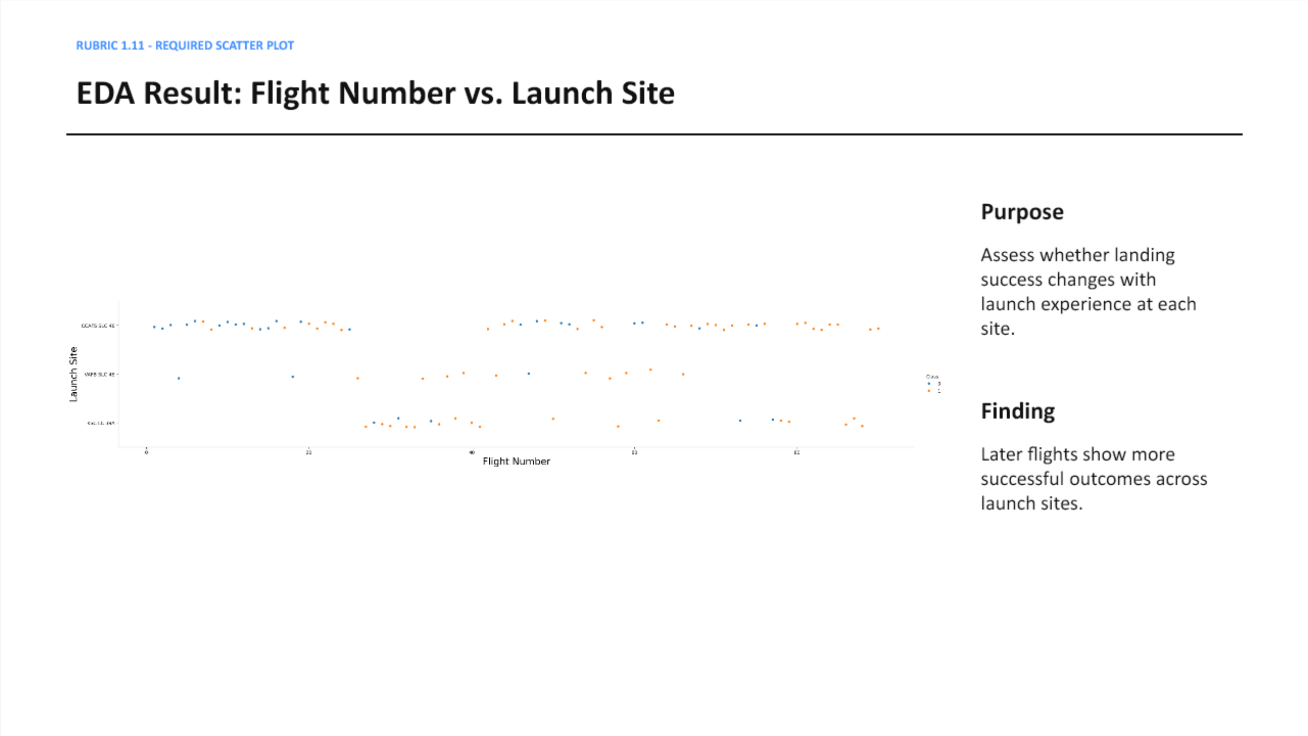


Supporting visual from the completed analysis

EDA Result: Flight Number vs. Launch Site

Evidence and interpretation

This scatter plot compares flight number and launch site, with point color representing landing outcome. Successful outcomes become more frequent in later flights across multiple sites, suggesting operational learning and maturation over time.

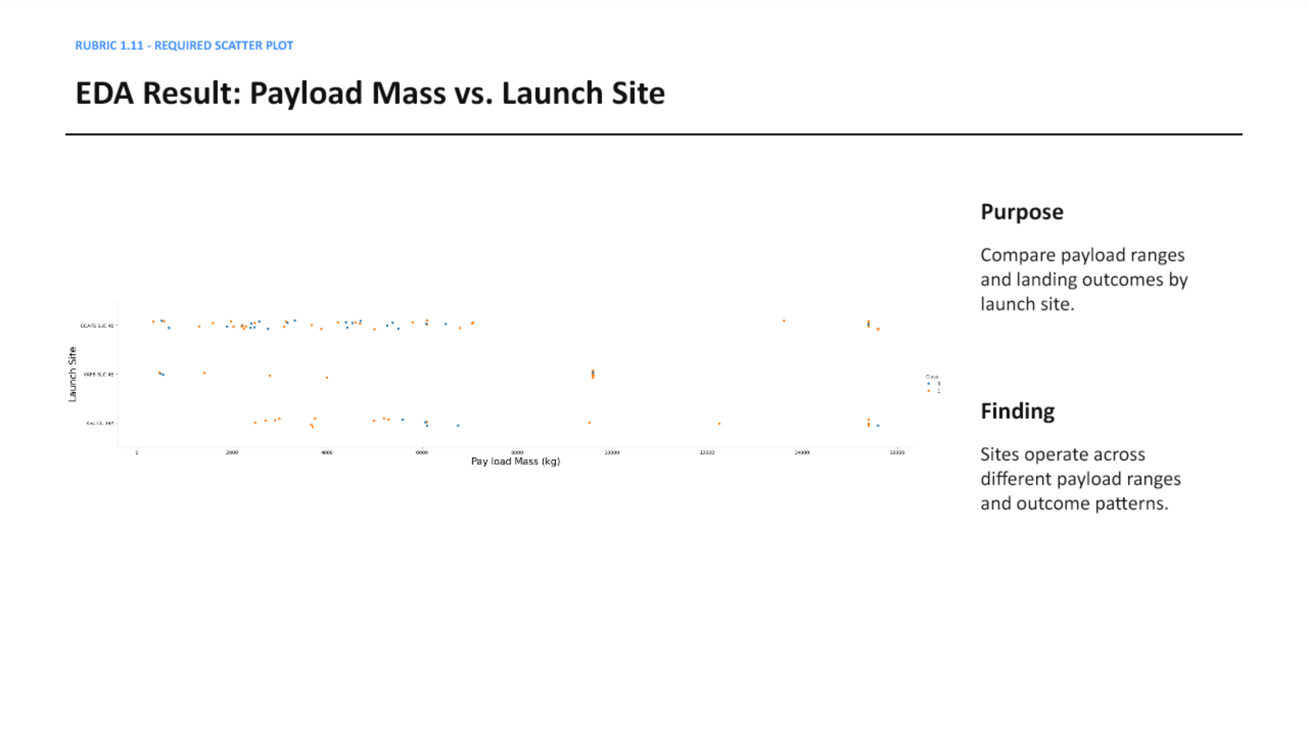


Supporting visual from the completed analysis

EDA Result: Payload Mass vs. Launch Site

Evidence and interpretation

This scatter plot compares payload mass across launch sites and distinguishes successful and failed landings. Launch sites operate over different payload ranges, so payload and site effects should be evaluated jointly rather than independently.

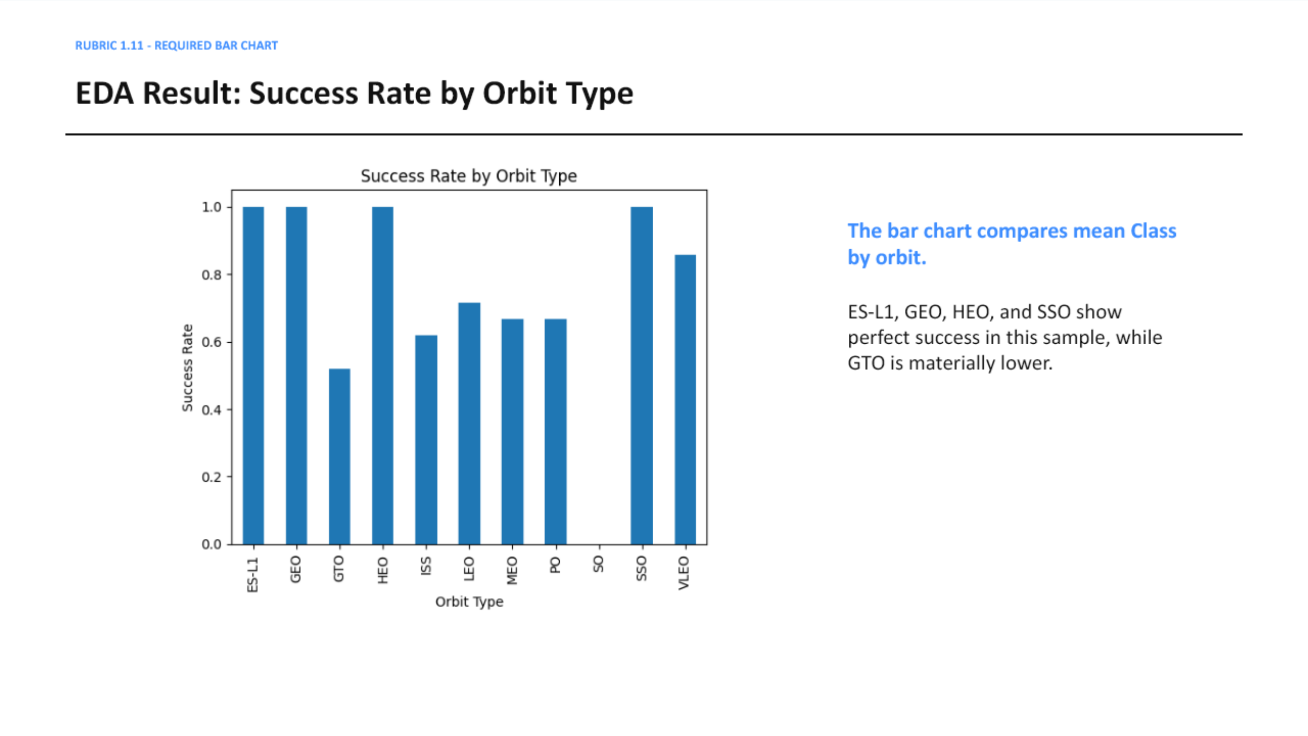


Supporting visual from the completed analysis

EDA Result: Success Rate by Orbit Type

Evidence and interpretation

The bar chart compares average Class by orbit type. ES-L1, GEO, HEO, and SSO show perfect success in this sample, while GTO has materially lower success. Orbit profile is therefore a useful predictive feature.

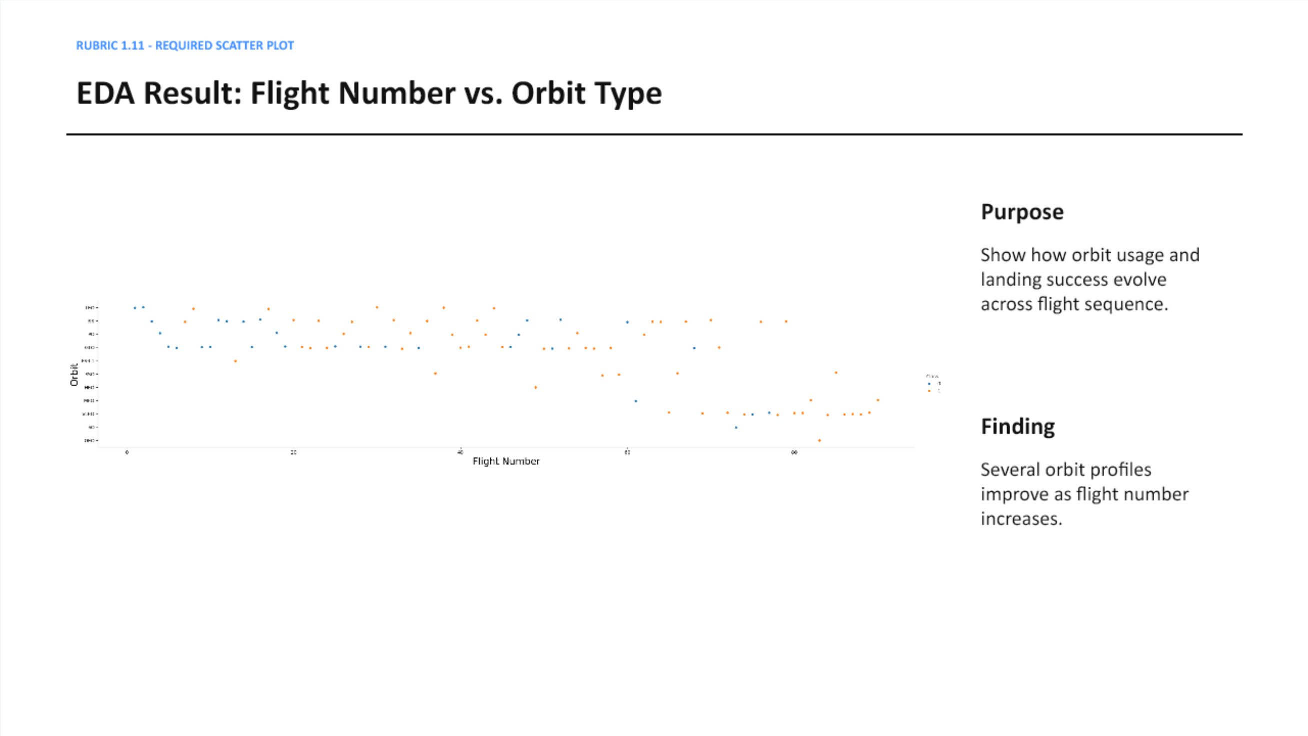


Supporting visual from the completed analysis

EDA Result: Flight Number vs. Orbit Type

Evidence and interpretation

This scatter plot compares flight sequence and orbit type. Several orbit categories show more successful outcomes at higher flight numbers, supporting the broader pattern of improvement as Falcon 9 operations matured.

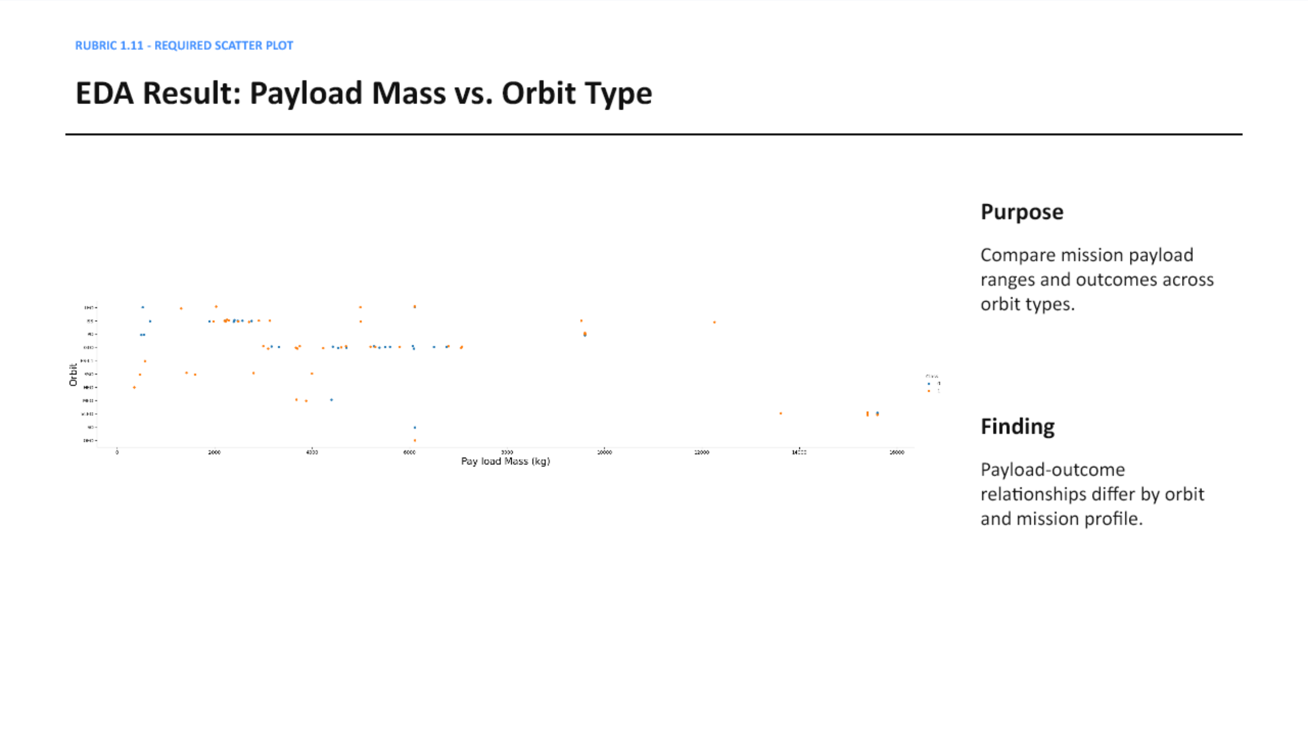


Supporting visual from the completed analysis

EDA Result: Payload Mass vs. Orbit Type

Evidence and interpretation

This plot compares payload mass and landing outcome within each orbit category. The relationship varies by orbit and mission profile, indicating that payload effects interact with destination orbit rather than following one universal threshold.

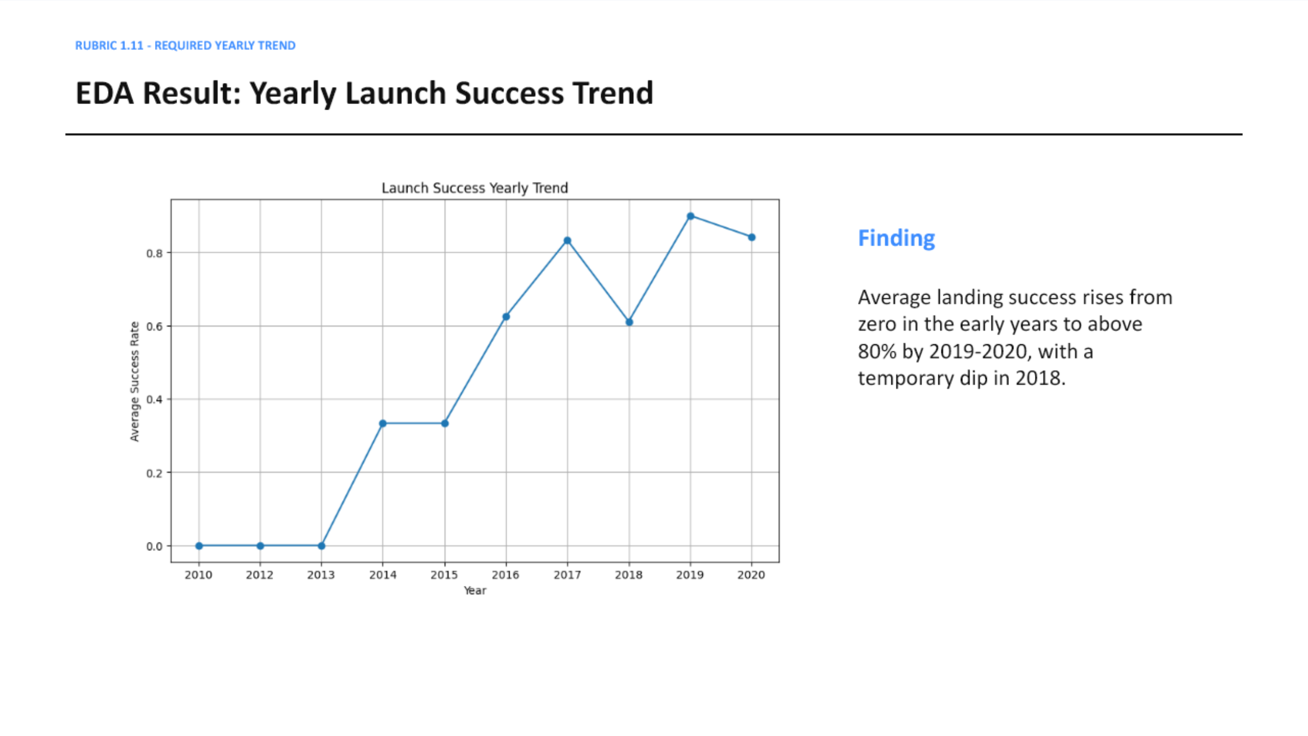


Supporting visual from the completed analysis

EDA Result: Yearly Launch Success Trend

Evidence and interpretation

Average landing success rises from near zero in the early years to above 80% by 2019-2020, with a temporary decline in 2018. The overall trend provides strong evidence of technological and operational improvement.



Supporting visual from the completed analysis

EDA with SQL Results: Launch Sites and Payloads

Evidence and interpretation

SQL returned four unique launch sites and four CCA records in the requested date window. NASA CRS payload totals 28,596 kg; average F9 v1.1 payload is 4,256.4 kg; and twelve booster records share the maximum payload.

RUBRIC 1.12 - SITES AND PAYLOAD ANALYSIS

EDA with SQL Results: Launch Sites and Payloads

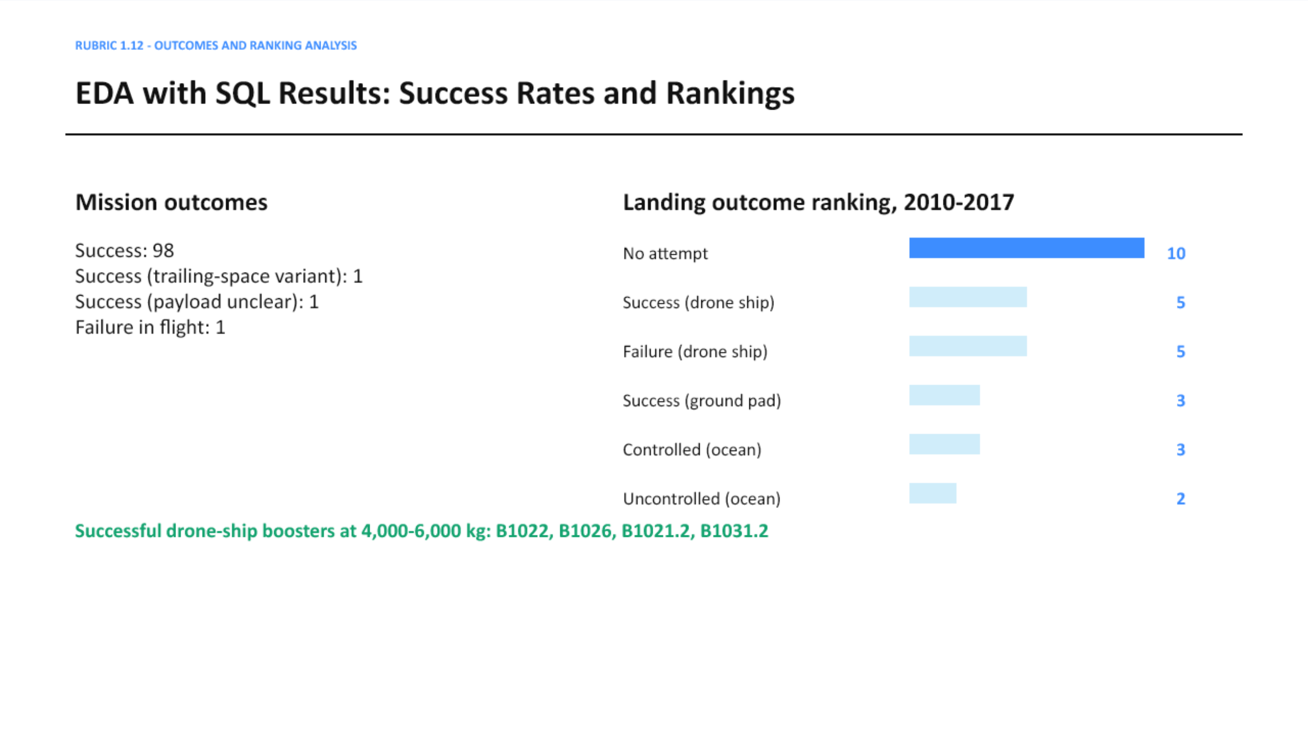
Unique launch sites	CCAFS LC-40; CCAFS SLC-40; KSC LC-39A; VAFB SLC-4E
CCA launch records	First five requested records returned
NASA CRS total payload	45,596 kg
F9 v1.1 average payload	2,928.4 kg
Maximum-payload boosters	12 Falcon 9 Block 5 booster records

Supporting visual from the completed analysis

EDA with SQL Results: Success Rates and Rankings

Evidence and interpretation

SQL aggregates summarize mission outcomes and rank landing outcomes from 2010-2017. Successful drone-ship boosters in the 4,000-6,000 kg payload interval include B1022, B1026, B1021.2, and B1031.2.

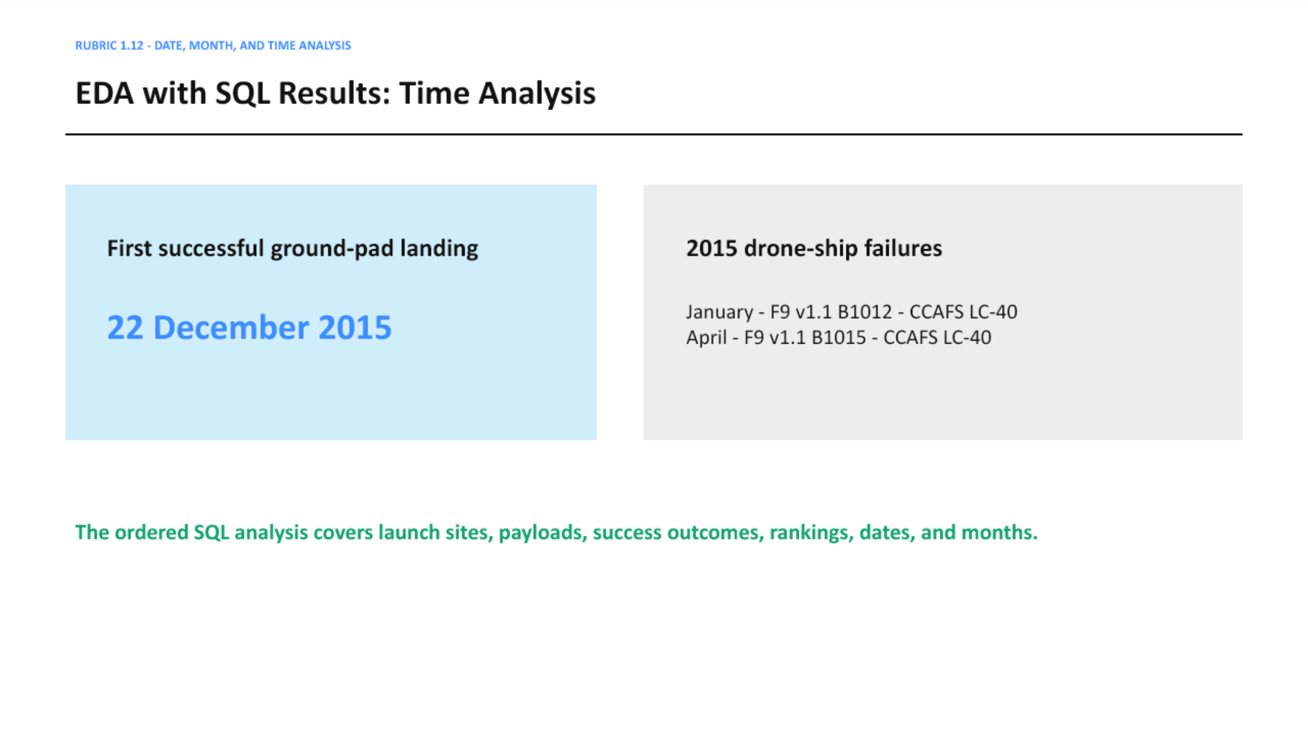


Supporting visual from the completed analysis

EDA with SQL Results: Time Analysis

Evidence and interpretation

The first successful ground-pad landing occurred on 22 December 2015. Monthly queries also identify the 2015 drone-ship failures and their associated booster and launch-site records.

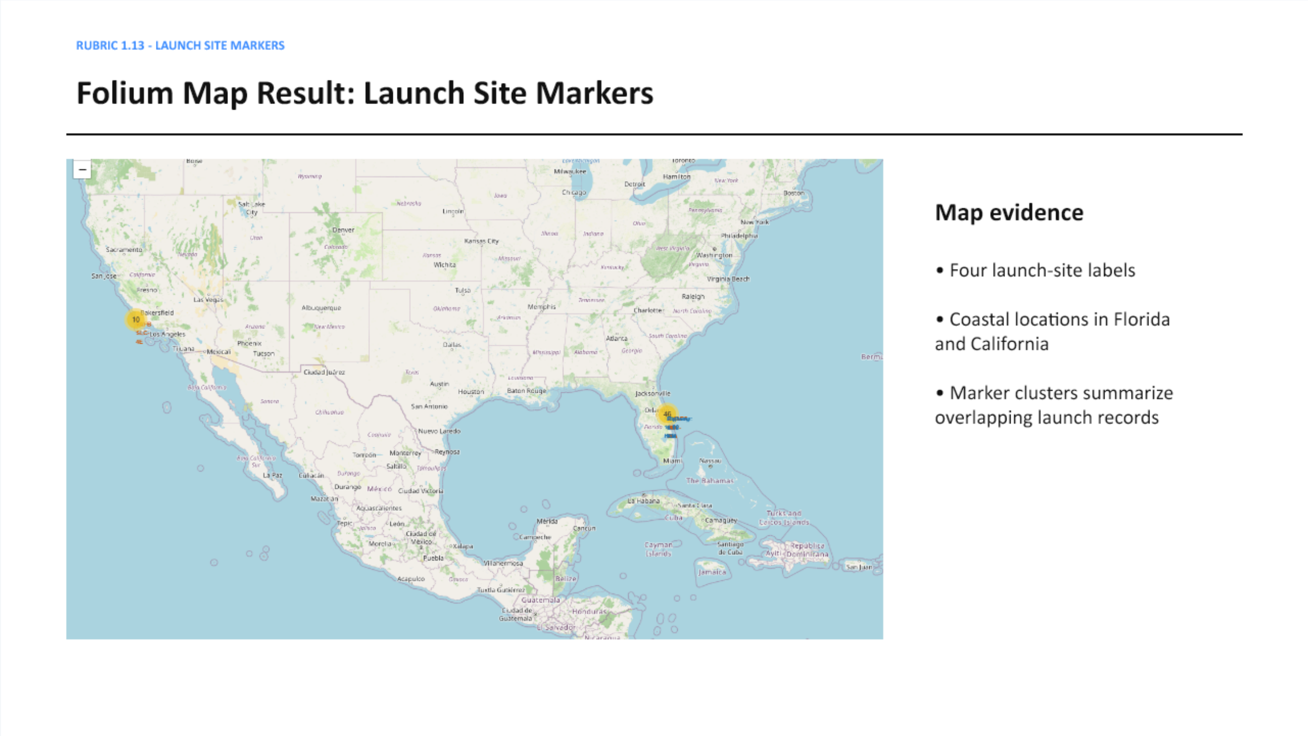


Supporting visual from the completed analysis

Folium Map Result: Launch Site Markers

Evidence and interpretation

The national Folium map shows all four launch-site labels at their coordinates. Marker clustering summarizes overlapping launch records and supports comparison of the geographic distribution of operations.

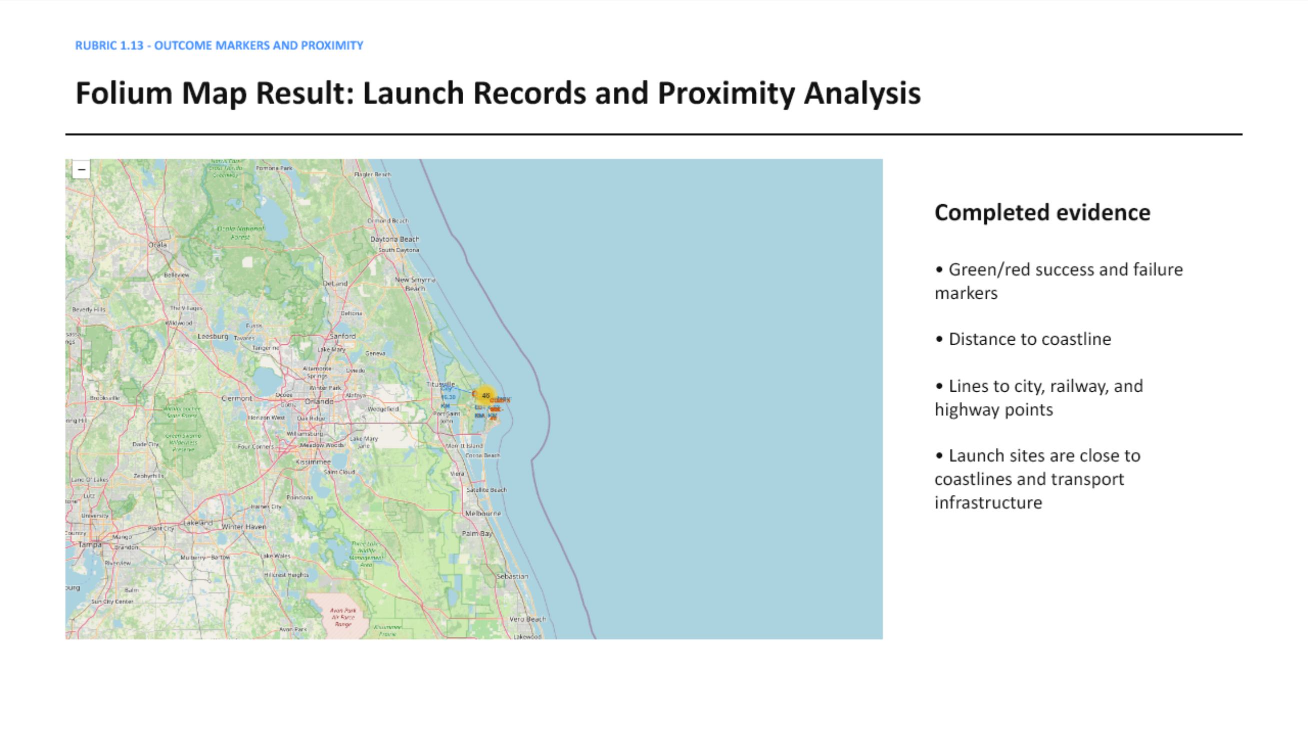


Supporting visual from the completed analysis

Folium Map Result: Launch Records and Proximity Analysis

Evidence and interpretation

Green and red markers distinguish successful and failed launch records. Distance lines show proximity to coastline, city, highway, and railway. The sites are close to the coast and transportation infrastructure, supporting range safety and logistics.

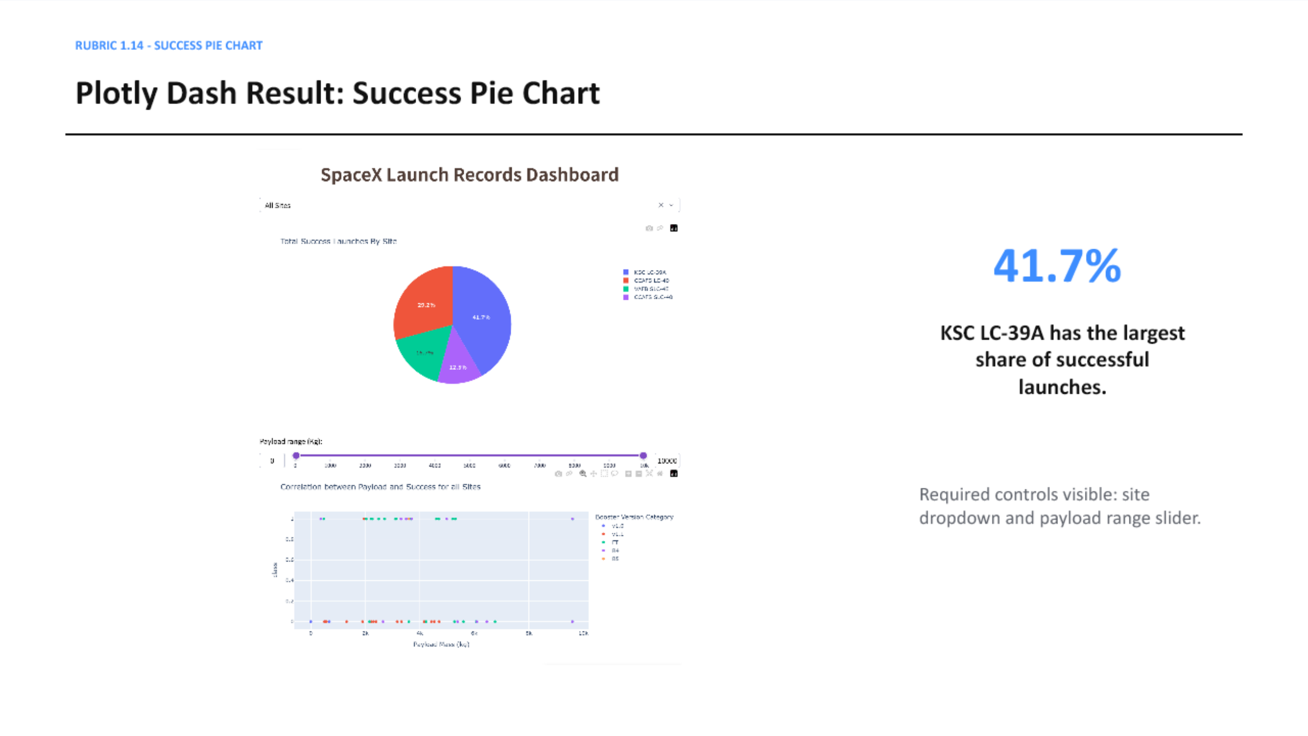


Supporting visual from the completed analysis

Plotly Dash Result: Success Pie Chart

Evidence and interpretation

The all-sites Dash pie chart reports the contribution of each site to successful launches. KSC LC-39A has the largest share at 41.7%. The dropdown and payload slider demonstrate interactive filtering controls.

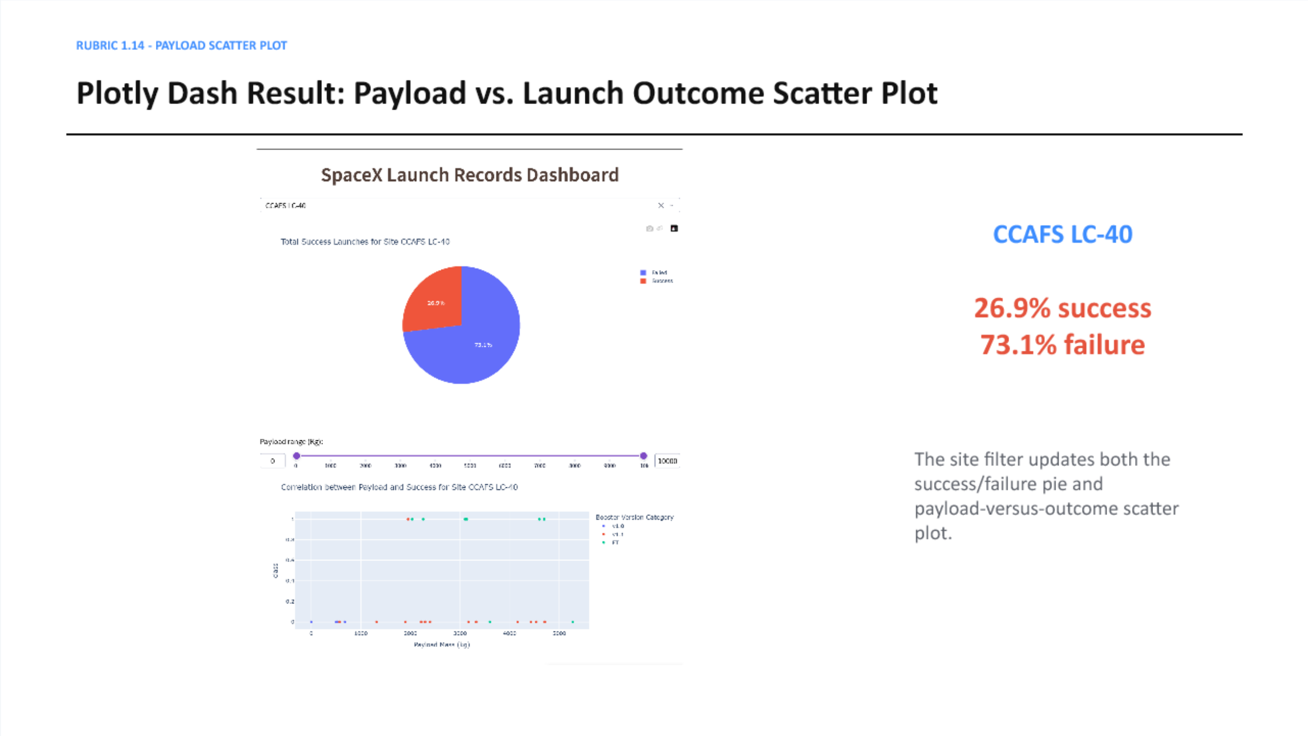


Supporting visual from the completed analysis

Plotly Dash Result: Payload vs. Launch Outcome Scatter Plot

Evidence and interpretation

Selecting CCAFS LC-40 updates both Dash charts. The site records show 26.9% successful launches and 73.1% failed launches, while the payload-outcome scatter plot responds to the selected site and payload interval.



Supporting visual from the completed analysis

Predictive Analysis: Multiple Models and Evaluation Results

Evidence and interpretation

GridSearchCV tuned logistic regression, support vector machine, decision tree, and K-nearest neighbors. Every classifier achieved 83.3% test accuracy. The decision tree achieved the strongest cross-validation score at 92.9%.

RUBRIC 1.15 - MODEL COMPARISON

Predictive Analysis: Multiple Models and Evaluation Results

Model	CV accuracy	Test accuracy
Logistic Regression	84.6%	83.3%
Support Vector Machine	84.8%	83.3%
Decision Tree	92.9%	83.3%
K-Nearest Neighbors	84.8%	83.3%

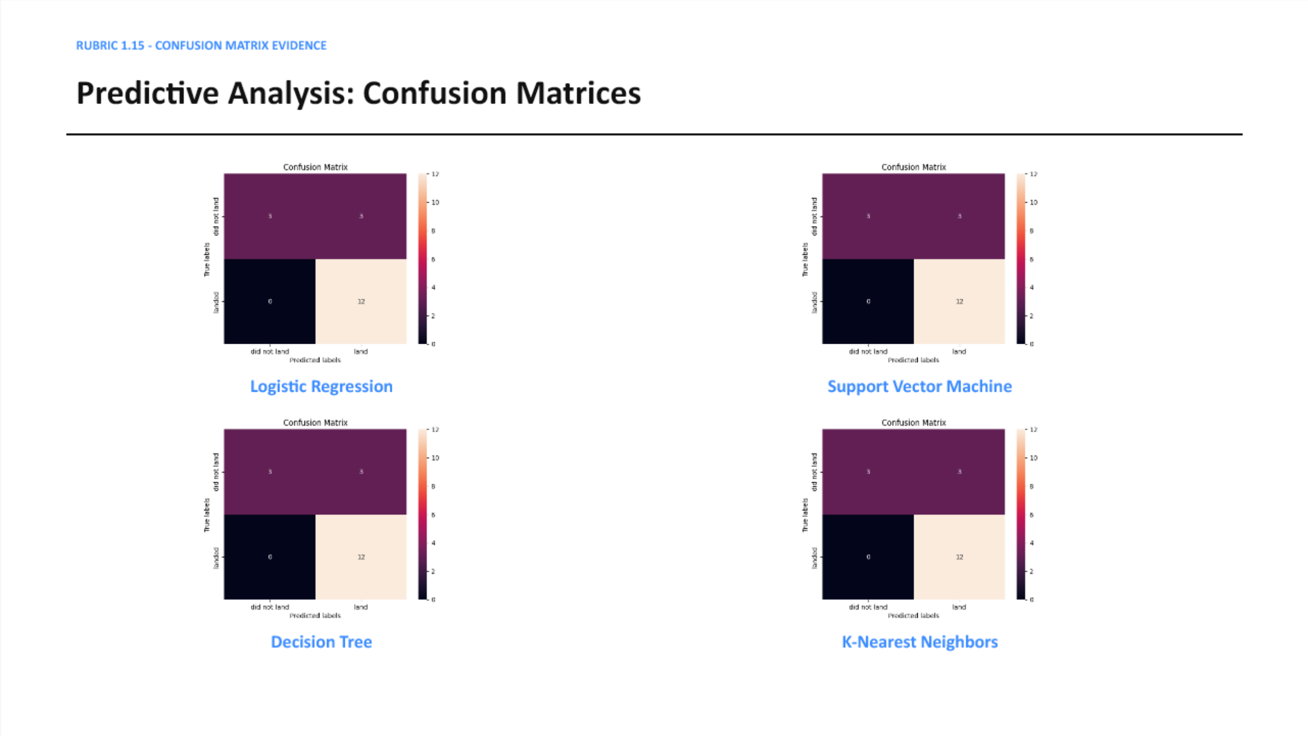
All models were tuned with GridSearchCV; the test set contains 18 launches.

Supporting visual from the completed analysis

Predictive Analysis: Confusion Matrices

Evidence and interpretation

Four confusion matrices display true positives, true negatives, false positives, and false negatives for logistic regression, SVM, decision tree, and KNN. This comparison supplements accuracy by showing the type and distribution of classification errors.

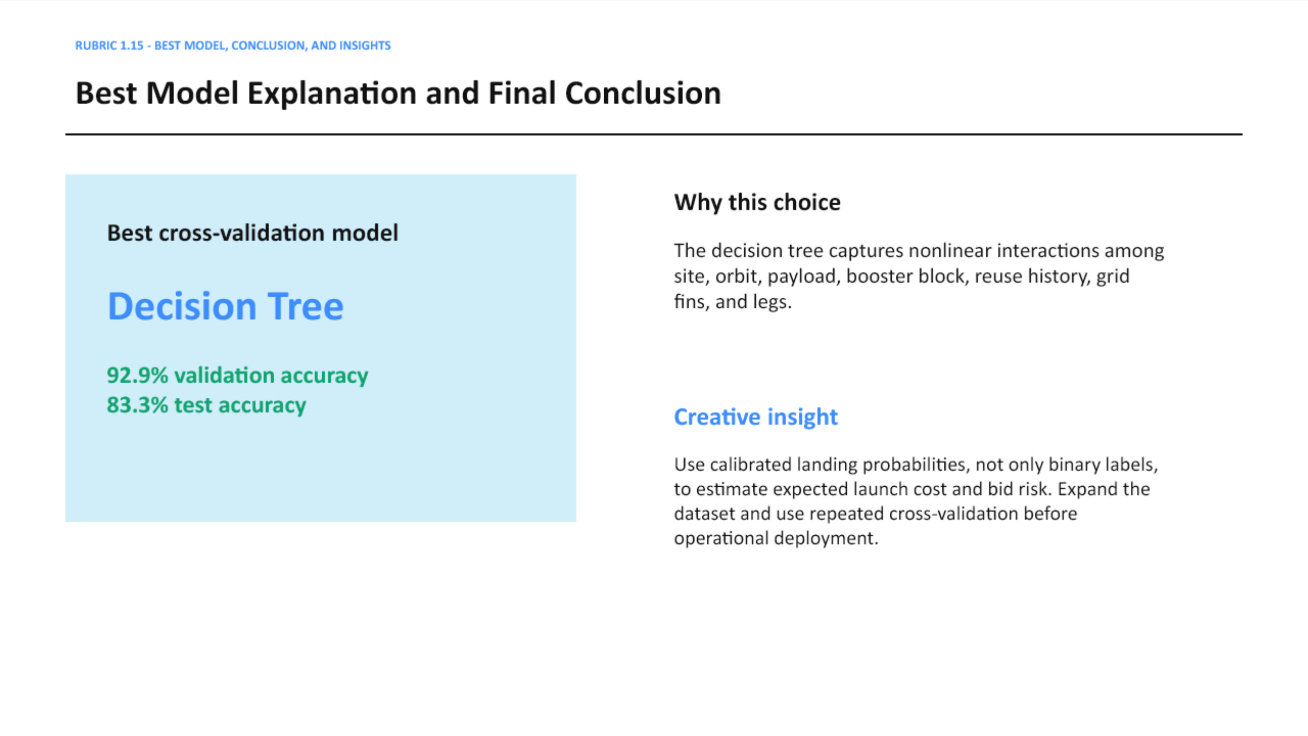


Supporting visual from the completed analysis

Best Model Explanation and Final Conclusion

Evidence and interpretation

The decision tree is selected because it combines the best cross-validation performance with interpretable nonlinear feature interactions. The model should support, not replace, engineering judgment. Future work should monitor drift, add weather and pad-condition variables, expand the data, and validate on newer launches. A key insight is that payload, orbit, site, and flight maturity interact rather than acting independently.



Supporting visual from the completed analysis

Rubric Evidence Index in Required Order

Evidence and interpretation

The evidence index maps criteria 1.1-1.15 to the corresponding pages. It confirms the repository URL, completed PDF, executive summary, introduction, collection and wrangling, EDA and SQL methods and results, Folium, Dash, predictive evaluation, conclusion, creativity, and innovative insight.

FINAL QUALITY CHECK - RUBRIC 1.1 THROUGH 1.15

Rubric Evidence Index in Required Order

1.1	GitHub URL	Slide 2	1.9	EDA SQL summary	Slide 9
1.2	Completed presentation PDF	Slide 1-27	1.10	Interactive summary	Slide 10
1.3	Executive Summary	Slide 3	1.11	EDA result slides	Slide 11-16
1.4	Introduction	Slide 4	1.12	SQL result slides	Slide 17-19
1.5	SpaceX API	Slide 5	1.13	Folium result slides	Slide 20-21
1.6	Web Scraping	Slide 6	1.14	Plotly Dash result slides	Slide 22-23
1.7	Data Wrangling	Slide 7	1.15	Predictive Analysis	Slide 24-26
1.8	EDA Visualization summary	Slide 8			

Supporting visual from the completed analysis